

Review Article

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PII: S2589-238X(26)00058-6

DOI: <https://doi.org/10.1016/j.hest.2026.08.004>

Reference: HEST 295

To appear in: *Hemorrhagic Stroke*

Received Date: 11 March 2026

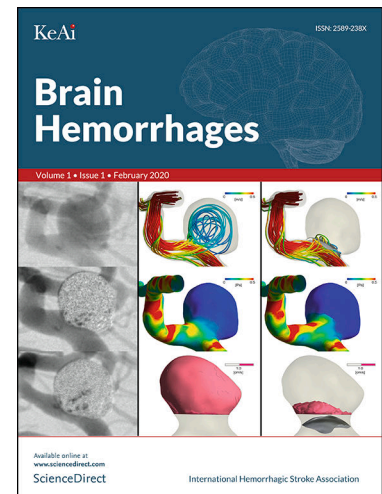
Revised Date: 4 August 2026

Accepted Date: 20 August 2026

Please cite this article as: R. Zhang, L. Cao, V.W. Yong, L. Yang, M. Xue, Decoding the spatiotemporal pleiotropy of CCL2-CCR2 signaling to guide staged therapies in CNS injury, *Hemorrhagic Stroke* (2026), doi: <https://doi.org/10.1016/j.hest.2026.08.004>

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Decoding the spatiotemporal pleiotropy of CCL2-CCR2 signaling to guide staged therapies in CNS injury¹

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Short title: CCL2-CCR2 signaling and CNS injury

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**CCL2-CCR2 Signaling in Intracerebral Hemorrhage and Other Acute
Central Nervous System Injuries: Spatiotemporal Mechanisms and
Staged Immunomodulation**

Abstract

Objective (Background): Intracerebral hemorrhage (ICH) and other acute central nervous system (CNS) injuries share a devastating pathological hallmark: neuroinflammation. Within this complex immune network, the CCL2-CCR2 signaling axis emerges as a critical driver connecting local tissue damage to systemic immune mobilization. However, the translational stagnation of CCL2-CCR2-targeted therapies—and the failure of broad-spectrum CCR2 blockade in non-CNS clinical trials—highlights that such static inhibition fundamentally ignores the dynamic, stage-specific shifting of myeloid phenotypes.

Methods: To address this bottleneck, our review delineates the spatiotemporal evolution of CCL2-CCR2 signaling, anchored primarily in hemorrhagic stroke pathophysiology, while utilizing ischemic, traumatic, and spinal cord injuries as comparative models.

Results: Following acute hemorrhagic or ischemic insults, rapidly upregulated CCL2 recruits peripheral monocytes, exacerbating secondary damage by promoting barrier disruption, oxidative stress, and iron-induced ferroptosis. Furthermore, we evaluate emerging precision-targeted interventions designed to bypass current limitations, including allosteric biased modulators, biomimetic nanodelivery platforms, and targeted transcriptional silencing.

Conclusion: Ultimately, we propose that dynamically staged immunomodulatory strategies are required to effectively balance early neuroprotection with late-stage hematoma clearance and tissue regeneration.

Funding: The authors' research on cerebrovascular diseases was supported by grants from the Henan Province Medical Science and Technology Research Program (SBGJ202403031); Natural Science Foundation of Henan (262300421603); Postdoctoral Fellowship Program of China Postdoctoral Science Foundation (GZC20232401); Hunan Provincial Natural Science Foundation (2023JJ40572); National Natural Science Foundation of China (U25A2065, W2541024); Key Advantageous Discipline Construction Project of Guizhou Provincial Health Commission (2023, Emergency Department); Guizhou Provincial Natural Science Foundation (Qiankehe Foundation MS[2025]432); National Natural Science Foundation of China (NSFC) Regional Fund Cultivation Program Project of the Affiliated Hospital of Guizhou Medical University (gyfynsfc[2023]-47).

Keywords: CCL2-CCR2 axis; Neuroinflammation; Monocyte chemoattractant protein-1; Intracerebral hemorrhage; Blood-brain barrier; Monocyte-derived macrophages; Secondary brain injury; Spatiotemporal immunomodulation

1. Introduction

Acute central nervous system (CNS) injuries encompass a spectrum of devastating physiochemical insults. While ischemic stroke, traumatic brain injury (TBI), and spinal cord injury (SCI) share common secondary inflammatory cascades, hemorrhagic injuries—particularly intracerebral hemorrhage (ICH) and subarachnoid hemorrhage (SAH)—present uniquely severe pathological challenges. These challenges are characterized by the mass effect of extravasated blood and the extreme neurotoxicity of erythrocyte degradation products. Despite their disparate initial triggers, these diverse pathological processes ultimately converge on a highly conserved core mechanism: the decompensation and collapse of neuroimmune homeostasis.¹⁻³ The impairment of blood-brain barrier integrity initiates this cascade by compromising the immune-privileged environment of the CNS. This disruption exposes the neural parenchyma to peripheral inflammatory infiltration.⁴ The subsequently activated neuroimmune network exhibits remarkable spatiotemporal pleiotropy and is characterized by resident glial polarization alongside transendothelial leukocyte recruitment. Early inflammatory responses exacerbate secondary neurotoxicity and tissue necrosis. In contrast, subacute and chronic immune activities serve as essential drivers for clearing pathological debris and initiating tissue remodeling.⁵⁻⁸

Within this highly intertwined immune network, the C-C motif chemokine ligand 2 (CCL2) and its cognate receptor CCR2 emerge as the pivotal molecular target coupling peripheral immune mobilization with local CNS pathology.⁶ Following acute injury, damage-associated molecular patterns (DAMPs) stimulate resident glia via Toll-like receptor 4/NF- κ B signaling to synthesize inflammatory mediators.^{9,10} Studies demonstrate that local CCL2 release exhibits pronounced niche-specificity.^{9,10} For example, extravasated thrombin in hemorrhagic lesions of SCI drives significant CCL2 upregulation in reactive astrocytes.¹⁰ This localized gradient counteracts bone marrow retention signals and directs the targeted infiltration of CCR2⁺ monocytes into the CNS. Beyond classical chemotaxis, recent single-cell transcriptomic analyses reveal that CCL2 shapes the pro-inflammatory polarization of infiltrating macrophages.¹¹⁻¹³ This polarization prompts the secretion of cytotoxic factors (e.g., tumor necrosis factor- α , interleukin-1 β) and reactive oxygen species (ROS). Consequently, this process exacerbates parenchymal destruction following ischemic or hemorrhagic CNS injuries.^{11,12} It also exerts non-canonical functions by directly inducing neuronal excitotoxicity and increasing endothelial permeability.^{14,15}

Given its central role in neuroinflammation, the CCL2-CCR2 pathway represents a highly promising target for clinical translation. Nevertheless, its pharmacological application is fundamentally hindered by functional pleiotropy and time-dependency.

Experimental and clinical evidence demonstrates that early blockade of CCL2 signaling mitigates secondary neural damage. However, non-selective inhibition of CCR2-mediated myeloid recruitment during later stages severely impairs the clearance of pathological products and tissue repair. Analogous evidence from pulmonary disease further supports this concept, where observations in patients with human CCR2 genetic deficiency highlight the absolute necessity of CCR2-dependent monocyte recruitment for essential tissue clearance functions.^{16,17} To overcome this translational bottleneck, this review delineates the spatiotemporal pathological mechanisms of the CCL2-CCR2 axis across prototypical CNS injuries and delayed neurodegeneration. By evaluating the critical nodes of monocyte recruitment, dynamic myeloid state transitions, and neuro-glial interactions, we aim to provide a scientific rationale for phase-specific immunomodulatory strategies. These strategies are designed to balance early neuroprotection with late-stage tissue regeneration.

2. Time-dependent and disease-dependent roles of CCL2-CCR2 signaling in acute CNS injury

The biological effects of CCL2-CCR2 signaling after acute CNS injury vary with post-injury timing, disease type, and the local tissue microenvironment. Across hemorrhagic, ischemic, traumatic, and spinal cord injuries, this axis has been implicated in chemokine induction, peripheral myeloid-cell recruitment, neurovascular injury, clearance responses, and tissue remodeling. Therefore, CCL2-CCR2 signaling should be discussed as a time- and disease-dependent pathway rather than as a uniformly detrimental inflammatory cascade.

2.1 Temporal changes in CCL2-CCR2 signaling after acute CNS injury

Following an acute primary CNS insult, the pathological microenvironment evolves through sequential processes including danger-signal release, chemokine induction, peripheral immune-cell recruitment, inflammatory amplification, clearance responses, and tissue remodeling. Phase-resolved profiling in SCI has shown that cytokine and chemokine programs vary across post-injury time points, supporting the need to interpret chemokine-targeted interventions in relation to disease stage.¹⁸ In ICH, chemokine release after hematoma formation links local inflammatory activation to peripheral immune-cell chemotaxis across the disrupted blood-brain barrier (BBB), indicating that a time-dependent framework is also relevant to hemorrhagic brain injury (Table 1).¹⁹

During the early post-injury period, damage-associated molecular patterns and blood-derived components rapidly activate resident glial and vascular-associated cells. In experimental SCI, thrombin induces astrocytic CCL2 production through ERK/JNK/NF- κ B signaling, and this early CCL2 response precedes later CCL5

expression and sustained macrophage/microglial infiltration.¹⁰ Although this evidence is derived from SCI, it is relevant to hemorrhagic CNS injury because blood-derived components are also central elements of the ICH microenvironment. In human ICH, circulating CCL2 has been associated with poor functional outcome, supporting the clinical relevance of CCL2-related inflammatory signaling in hemorrhagic brain injury.²⁰

In the first days after ICH, the established CCL2-CCR2 axis promotes recruitment of peripheral CCR2-positive inflammatory monocytes into injured brain tissue. Experimental ICH studies show that blood-derived CCR2⁺Ly6C^{hi} inflammatory monocytes enter the brain, outnumber other leukocyte subsets, produce tumor necrosis factor, and exacerbate acute neurological disability.²¹ Similar early peripheral monocyte recruitment has also been described in ischemic stroke, where monocytes infiltrate lesions within 24 h, peak between 3 and 7 days, and subsequently differentiate into macrophages.²² In ICH, oxidative stress and NLRP3 inflammasome activation may further amplify secondary brain injury.²³ CCL2-CCR2 signaling has also been linked to BBB injury after acute ICH, as S1PR3-mediated BBB disruption was reported to involve the CCL2-CCR2 axis.²⁴

As injury progresses beyond the early inflammatory period, recruited myeloid cells may contribute to clearance and remodeling responses. Single-cell transcriptomic analysis after brain injury has identified immunoregulatory and neutrophil-like monocyte subsets, indicating that post-injury monocyte responses are heterogeneous rather than fixed in a single inflammatory state.²⁵ In ICH, CCL2 or CCR2 deficiency reduces early hematoma volume but delays subsequent hematoma recovery, suggesting that this axis participates in both hematoma progression and hematoma clearance.²⁶ Macrophage-mediated erythrophagocytosis further supports the importance of clearance-associated myeloid functions after ICH.²⁷ During later recovery stages, microglia-astrocyte interactions may contribute to neuroprotective glial scar formation after ICH, whereas TBI studies suggest that CCL2/CCR2-related responses may persist beyond the immediate injury period and do not map linearly onto all chronic pathological endpoints.^{16,28,29}

2.2 Disease-dependent and model-dependent differences in CCL2-CCR2 function

The evidence summarized above indicates that CCL2-CCR2 function depends on the biological process being assessed. In ICH, early CCR2⁺Ly6C^{hi} monocyte recruitment is associated with neurological worsening.²¹ However, CCL2 or CCR2 deficiency also delays later hematoma recovery, suggesting involvement of the same pathway in clearance-related processes.²⁶ This interpretation is consistent with evidence that macrophage erythrophagocytosis promotes hematoma resolution and neurological recovery after experimental ICH.²⁷ Thus, divergent conclusions may reflect differences in endpoints, such as acute neurological deficit, hematoma volume, hematoma

resolution, or long-term functional recovery.

Disease-specific pathological processes also affect how CCL2-CCR2-related myeloid responses should be interpreted. In ICH, blood-derived products, erythrocyte degradation, iron handling, and hematoma clearance are central features of the lesion environment. In ischemic stroke, CCR2-dependent monocytes/macrophages can aggravate acute injury but also support later functional recovery, indicating that the same myeloid lineage may have different effects across disease stages.³⁰ In SCI, exposure to myelin debris alters macrophage phenotype and function, suggesting that lesion-derived substrates can shape macrophage behavior.³¹ In TBI, *Ccr2* deletion dissociates lesion cavity size from tau pathology, suggesting that CCR2-related monocyte recruitment and chronic neurodegenerative pathology are not necessarily coupled in a simple manner.¹⁶

Differences in experimental models and intervention methods further limit direct comparison across studies. In ICH research, collagenase-induced and autologous blood injection models differ in vascular disruption, hematoma expansion, perihematomal injury, and inflammatory activation, which may influence conclusions regarding CCL2-CCR2 function.³² Moreover, genetic *Ccl2/Ccr2* deletion, pharmacological CCR2 antagonism, FROUNT inhibition, CCL2-siRNA delivery, and chemokine-decoy nanoparticles act at different levels of the pathway and should not be regarded as equivalent interventions. Cell-identification strategies also differ among studies; conventional markers may not fully distinguish infiltrating monocyte-derived macrophages from resident microglia, whereas single-cell transcriptomic studies show that post-injury monocytes include heterogeneous immunoregulatory and neutrophil-like subsets.²⁵ These considerations indicate that CCL2-CCR2-targeted strategies should account for disease type, intervention timing, delivery approach, and myeloid-cell state.

Furthermore, a major limitation hindering the clinical translation of current preclinical CNS injury studies is the overwhelming reliance on young, otherwise healthy rodent models. These models fundamentally fail to recapitulate the advanced age structure and complex comorbidity burden intrinsic to human stroke and hemorrhagic CNS injury populations.^{33,34} Advanced age and pre-existing vascular or metabolic comorbidities can profoundly alter baseline neurovascular homeostasis and inflammatory priming. This includes exacerbating endothelial senescence, heightening BBB vulnerability, and driving diabetes-associated barrier dysfunction following stroke or ICH.³⁵⁻³⁷ For instance, in experimental models of hemorrhagic stroke, concurrent type I diabetes has been shown to exacerbate post-injury cognitive impairment by significantly amplifying local oxidative stress and neuroinflammation, underscoring the absolute necessity of considering the metabolic disease background when interpreting post-ICH inflammatory dynamics.³⁸ Therefore, to rigorously bridge the translational gap, future studies evaluating CCL2-CCR2-targeted interventions must incorporate aged animals and clinically relevant comorbidity models—such as hypertension- or diabetes-associated cerebrovascular injury—and rigorously validate

key findings across these diverse model systems before advancing toward clinical trials.^{33,34}

3. Early CCL2 production and immune cell mobilization following injury

3.1 Upregulation of CCL2 expression triggered by DAMPs

The initiation of neuroinflammation following CNS injury is primarily triggered by DAMPs. Necrotic cells or extravasated blood components release these molecules into the microenvironment.^{9,10,39} In pathological conditions such as ICH and SCI, vascular rupture allows heme and thrombin to enter the neural parenchyma. These substances subsequently function as potent DAMPs. Thrombin is a serine protease that cleaves PAR-1 on the surface of astrocytes. This cleavage triggers the downstream phosphorylation of ERK1/2 and p38 MAPK. Consequently, this signaling cascade activates the NF- κ B pathway and promotes the transcriptional upregulation of the *Ccl2* gene.¹⁰ Concurrently, nuclear proteins such as HMGN1 are released into the extracellular space. They serve as ligands for TLR4 on microglia and astrocytes.⁹ This interaction initiates a MyD88-dependent signaling cascade.⁹ The cascade then promotes the nuclear translocation of NF- κ B p65 and the subsequent synthesis of pro-inflammatory chemokines including CCL2.⁹ Additionally, ATP released from damaged neurons activates P2X7 receptors on microglia. This activation further increases CCL2 production via the NLRP3 inflammasome pathway.³⁹ These combined pathways contribute to the rapid and localized upregulation of CCL2 levels during the early acute phase. This upregulation establishes the initial chemotactic microenvironment for subsequent leukocyte recruitment.

3.2 Systemic recruitment of peripheral inflammatory monocytes

Locally upregulated CCL2 can enter the systemic circulation through the compromised vascular barrier or the cerebrospinal fluid drainage system. Elevated plasma CCL2 levels constitute a key factor in inducing emergency myelopoiesis.^{16,40,41} This process features the accelerated generation and mobilization of myeloid cells within the bone marrow and spleen. The egress of inflammatory monocytes from the bone marrow is primarily regulated by the CCL2-CCR2 signaling axis. These monocytes are phenotypically characterized as Ly6C^{hi}CCR2⁺ in mice and CD14⁺CD16⁺CCR2⁺ in humans.^{17,40} Under homeostatic conditions, monocytes are retained in the bone marrow via CXCR4-CXCL12 interactions. Elevated plasma CCL2 levels activate CCR2 on the monocyte surface. This activation antagonizes the bone marrow retention signals and triggers monocyte release into the bloodstream.¹⁶ The spleen serves as an important extramedullary reservoir alongside the bone marrow. It actively participates in

releasing monocytes following acute CNS injury. In models of ischemic stroke, CCR2-deficient animals exhibit attenuated splenic contraction and reduced Ly6C^{hi} monocyte release. This finding demonstrates the involvement of this receptor in coordinating the systemic immune response.⁴² Upon entering the circulation, mobilized Ly6C^{hi}CCR2⁺ monocytes migrate toward the CNS along the CCL2 concentration gradient. They eventually accumulate at the lesion site (Figure 1).⁴³

3.3 CCL2-mediated alterations in blood-brain barrier permeability

The infiltration of peripheral immune cells into the CNS parenchyma is typically accompanied by increased permeability of the BBB or blood-spinal cord barrier. Physical injury can directly cause barrier disruption.^{4,15} However, sustained alterations in permeability involve the participation of secondary biochemical signals. Studies indicate that CCL2 regulates the barrier function of endothelial cells through independent mechanisms beyond its classical chemotactic role. Brain microvascular endothelial cells (BMECs) express CCR2 on their surface. The binding of CCL2 to endothelial CCR2 induces the intracellular phosphorylation of p38 MAPK.^{4,15} This signal transduction leads to the downregulation and altered distribution of tight junction proteins such as occludin and claudin-5.^{4,15} Consequently, this process compromises paracellular barrier function.^{4,15} Furthermore, the endfeet of perivascular astrocytes act as a major local source of CCL2. They likely participate in this process of permeability alteration. In ischemic stroke models, Slc4a4-mediated intracellular alkalinization in astrocytes promotes CCL2 secretion. The secreted CCL2 then acts on endothelial CCR2 to reduce tight junction integrity via a NO-dependent mechanism.¹⁵ Similarly, blocking CCR2 signaling in subarachnoid hemorrhage models mitigates BBB damage by inhibiting the p38 MAPK pathway. This observation further supports the vital role of this signaling axis in regulating endothelial barrier function (Figure 2).²

4. CCL2-mediated cellular microenvironment remodeling

4.1 Recruitment and differentiation of peripheral monocytes

Increased BBB permeability allows peripheral immune cells to enter the parenchyma. This infiltration significantly alters the cellular composition at the injury site. The CCL2-mediated recruitment of Ly6C^{hi}CCR2⁺ monocytes plays a vital role in this process. Upon entering the CNS, these cells differentiate into MoDMs. Early studies struggled to distinguish infiltrating MoDMs from resident microglia due to overlapping surface markers. Recently, scRNA-seq technology has provided cellular atlases with much higher resolution.^{13,25} Studies show that infiltrating MoDMs accumulate in the core lesion area within days following acute injuries such as ischemic stroke or TBI.

These recruited cells exhibit distinct transcriptional profiles compared to resident microglia.²⁵ Resident microglia tend to express homeostasis-related or repair-related genes such as *P2ry12* and *Tmem119*.^{13,25} Conversely, recruited MoDMs typically display transcriptomes enriched in glycolytic metabolic genes and pro-inflammatory mediators.²⁵ In models of brain injury induced by myocardial infarction, the infiltration of CCR2⁺ monocytes correlates with white matter inflammation and cognitive dysfunction.⁴³ Depleting CCR2 effectively reduces this infiltration. Importantly, this early depletion exerts minimal impact on the resident glial populations.⁴³

4.2 CCL2-mediated activation of pro-inflammatory phenotypes

The CCL2-CCR2 signaling axis goes beyond classical chemotaxis. It also regulates the phenotype of myeloid cells within the damaged CNS. However, recent single-cell RNA sequencing (scRNA-seq) studies suggest that the traditional binary "M1/M2" classification may not fully capture the complexity of myeloid heterogeneity, as recruited monocytes and resident microglia appear to adopt more continuous, disease-associated transcriptional trajectories.^{13,25,44} Furthermore, since several recent transcriptomic atlases indicate that functional CCR2 expression on resident microglia is extremely low or absent, the observed microglial phenotypic shifts following CCL2-CCR2 modulation might be largely driven by secondary consequences of intense intercellular cross-talk rather than exclusively through direct CCL2 responses.^{10,13,25,28,44} Consequently, the terms "pro-inflammatory" and "reparative" are utilized here primarily to describe functional tendencies.

Upon receptor binding, CCL2 initiates intracellular signaling cascades. These cascades promote the shift of infiltrating monocytes and, potentially, resident microglia toward a pro-inflammatory phenotype. This transition is primarily mediated by the NF- κ B and MAPK signaling pathways. Specifically, CCL2 induces the phosphorylation and nuclear translocation of the NF- κ B p65 subunit. This event drives the transcription of pro-inflammatory genes including *Nos2*, *Ptgs2*, and *Il1b*.^{45,46} Furthermore, studies demonstrate that inhibiting the CCL2-CCR2 axis downregulates these pro-inflammatory markers. This inhibition is accompanied by the upregulation of reparative markers such as CD206 and arginase-1. This finding suggests a vital role for CCL2 in maintaining the pro-inflammatory polarized state.^{11,12}

Classically activated cells secrete cytokines including TNF- α , IL-1 β , and IL-6. These factors stimulate astrocytes and endothelial cells via paracrine signaling to produce additional CCL2. This interaction creates a positive feedback loop for inflammatory amplification.⁴⁷ Concurrently, polarized macrophages upregulate iNOS and NADPH oxidase. This upregulation leads to the accumulation of ROS and NO. Such oxidative stress increases the risk of lipid peroxidation and DNA damage in adjacent neurons and endothelial cells. This activation triggers GSDMD cleavage and IL-1 β release. Ultimately, these processes sustain the inflammatory state of the local

pathological microenvironment.⁴⁸⁻⁵⁰

4.3 Interactive signaling between astrocytes and macrophages

CCL2-mediated myeloid cell infiltration also affects the physiological functions of astrocytes. Astrocytes act as a major source of CCL2 during the acute phase of injury. However, they also respond to inflammatory mediators released by the recruited myeloid cells.¹⁰ Infiltrating M1-like macrophages and activated microglia secrete specific cytokines. These cytokines induce the transition of homeostatic astrocytes into a reactive (A1-like) phenotype.^{51,52} These reactive astrocytes typically exhibit the downregulation of synapse-formation genes and the upregulation of complement components. Consequently, their neurosupportive functions are compromised to varying degrees.⁵¹ Mechanistically, this phenotypic transition involves the activation of the JAK2/STAT3 and NF- κ B pathways within astrocytes. Pro-inflammatory cytokines induce the phosphorylation of STAT3 at the Tyr705 residue alongside the nuclear translocation of the NF- κ B p65 subunit.⁴⁵ Furthermore, Lcn2 secreted by pro-inflammatory macrophages participates in this process.⁵³ It acts on astrocytes to upregulate chemokine secretion.⁵³ This interaction helps maintain the positive feedback mechanism of inflammatory signaling. Within this reciprocal network, CCL2 initially secreted by damaged astrocytes recruits CCR2⁺ monocytes. These recruited cells differentiate into M1 macrophages and release TNF- α and Lcn2. These factors subsequently stimulate astrocytes via the NF- κ B pathway to synthesize additional CCL2.⁵⁴ This mutually amplifying signaling mechanism likely drives the persistent presence of inflammatory cells within the parenchyma. Consequently, it actively contributes to the process of chronic neurodegeneration (Figure 3).³⁹

5. Secondary injury mechanisms in neurons and the neural microenvironment

5.1 Non-canonical regulation of excitotoxicity and synapse loss

CCL2 is classically considered a leukocyte chemoattractant. However, studies show it also regulates neuronal excitability and synaptic transmission via neuronal CCR2 receptors.¹⁴ Mechanistically, CCL2-CCR2 signaling independently modulates two distinct classes of ionotropic glutamate receptors. On one hand, CCL2 binding promotes the surface trafficking of GluA1 subunits, thereby enhancing AMPA receptor-mediated excitatory synaptic transmission.¹⁴ On the other hand, CCL2 signaling simultaneously enhances NMDA receptor-mediated currents, involving subunits such as GluN2B.⁵⁵ The upregulation of surface receptor density and hyperactivity of both AMPA and NMDA receptors can trigger excessive calcium influx in postsynaptic neurons following glutamate stimulation. Consequently, this increases

the risk of intracellular calcium overload.⁵⁵ This calcium dyshomeostasis correlates with mitochondrial depolarization and ROS production. These events collectively increase neuronal susceptibility to excitotoxicity.⁵⁶ Concurrently, inflammatory signals within the microenvironment disrupt synaptic homeostasis. Local pro-inflammatory factors promote microglial activation and phagocytosis.⁵⁷ The deposition of complement components like C3 at synaptic terminals tags them for elimination.^{6,51} Activated microglia subsequently clear these tagged synapses via the CR3 pathway.^{6,51} This synapse loss significantly impairs neural circuit connectivity. It represents a potential pathological basis for cognitive and sensory dysfunctions in chronic CNS injuries.³

5.2 Oxidative stress and cell death susceptibility driven by inflammation

The neuroinflammatory microenvironment regulated by the CCL2-CCR2 axis interacts reciprocally with the local oxidative stress state.^{23,56} Mechanistic studies suggest that CCL2 receptor binding facilitates the assembly of the NOX2 complex in macrophages and microglia. This assembly promotes the generation of ROS such as superoxide anions.⁵⁸ ROS act as intracellular signaling molecules. They sustain the activation of MAPK and NF- κ B pathways to drive the transcription of pro-inflammatory factors.⁵⁶ Particularly in pathologies like ICH, ROS accumulation closely correlates with the hyperactivation of the NLRP3 inflammasome. This interaction creates a localized and mutually amplifying cycle of oxidative stress and inflammation,²³ which further orchestrates critical downstream metabolic and hypoxic signaling networks.⁵⁹ Concurrently, CCL2 signaling inhibits the nuclear translocation of Nrf2. This inhibition downregulates downstream antioxidant targets including HO-1 and GPX4.⁶⁰ Consequently, the cellular capacity to neutralize lipid hydroperoxides is significantly diminished. In hemorrhage-associated pathologies like ICH and SCI, the degradation of erythrocytes releases free ferrous ions (Fe^{2+}). This release exacerbates the local oxidative burden.⁶¹⁻⁶³ Under inflammatory conditions, the iron-catalyzed Fenton reaction readily generates highly toxic hydroxyl radicals.⁵⁸ These radicals attack polyunsaturated fatty acids in neuronal membranes. This attack generates lipid peroxides that actively drive ferroptotic cell death.⁵⁸ Furthermore, CCL2-induced inflammatory cytokines inhibit the System Xc⁻ antiporter. This inhibition leads to intracellular GSH depletion and further impairs GPX4 activity.⁶⁴ Therefore, the CCL2-CCR2 axis establishes a toxic feed-forward loop. In this loop, inflammation breeds oxidative stress. The ensuing oxidative stress conversely induces ferroptosis and secondary tissue damage (Figure 4).⁶⁴

5.3 Microenvironmental restrictions on myelin and axonal repair

Following CNS injury, the inflammatory cascade mediated by the CCL2-CCR2 axis shapes a complex microenvironment. This altered niche is closely associated with

oligodendrocyte dysfunction and axonal degeneration.⁶⁵⁻⁶⁷ In SCI models, the specific characteristics and dynamic evolution of local immune cell infiltration dictate the extent of secondary neural damage and subsequent remodeling capacity.⁶⁵ Experiments demonstrate that targeted interventions lowering CCL2 levels effectively mitigate local inflammation. This process likely protects myelin and neurons at the lesion site by disinhibiting survival signaling pathways such as PI3K/Akt.⁶⁸ Additionally, CCL2-driven microenvironmental changes further compromise axonal homeostasis by reducing BBB integrity. In brain injuries like ischemic stroke, the aberrant activation of this pathway induces BBB leakage. This leakage exposes vulnerable axons to the toxic effects of peripheral blood components and increases the risk of axonal oxidative damage.¹⁵ Notably, the role of chemokines in neural tissue repair exhibits pronounced tissue specificity. In PNS injury models, upregulated CCL2 at the lesion site recruits peripheral macrophages. Within the PNS environment, these macrophages actively clear myelin debris. This clearance is an essential prerequisite for promoting axonal regeneration.^{69,70} Conversely, within the CNS environment, therapeutically modulating the inflammatory microenvironment improves myelin repair following SCI.⁷¹ This highlights that a sustained pro-inflammatory microenvironment acts as a major barrier restricting CNS neural repair. Exploring strategies to ameliorate this inflammatory microenvironment is therefore crucial. It mitigates endogenous damage and serves as a fundamental prerequisite for optimizing future regenerative therapies such as neural stem cell transplantation.⁷²

6. Disease-specific pathological mechanisms of the CCL2-CCR2 axis

Although CCL2-CCR2-mediated monocyte recruitment is a shared feature of acute CNS injuries, the subsequent phenotypic differentiation and pathological functions of these myeloid cells are strictly dictated by the local injury microenvironment. Specifically, diverse primary insults—ranging from hematoma formation and ischemia-reperfusion to mechanical trauma and axonal shearing—generate distinct biochemical substrates. These distinct substrates directly determine whether infiltrating CCR2⁺ macrophages predominantly amplify neurotoxicity or transition towards debris clearance and tissue remodeling. Therefore, the biological role of the CCL2-CCR2 axis must be evaluated within the specific pathological context of each CNS disorder.

6.1 Intracerebral hemorrhage: erythrocyte degradation, iron overload, and ferroptosis

The pathophysiology of ICH is closely linked to the physical presence of the hematoma and the secondary toxicity of its degradation products.¹⁹ Initially, perihematomal astrocytes actively attempt to counteract vascular disruption and maintain neurovascular integrity through the secretion of specific secreted factors, such as LAMC1.⁷³ However, in the wake of erythrocyte lysis, hemoglobin-derived heme and

iron accumulate in perihematomal tissue, where iron overload readily promotes oxidative stress, lipid peroxidation, and ferroptosis-related neuronal injury.^{62,74} Rodent-based experimental studies further support ferroptosis as a strictly regulated neuronal death pathway driving hemorrhagic brain injury.⁶¹ Paralleling this oxidative cascade, RIPK3-dependent necroptosis has been mechanistically linked to MCP-1/CCL2-mediated inflammatory amplification after ICH.⁷⁵

Embedded within this toxic milieu, the early CCR2-dependent recruitment of murine Ly6C^{hi} inflammatory monocytes (analogous to human CD14⁺CD16⁺ classical monocytes) rapidly aggravates acute neurological injury.²¹ Complicating this narrative, however, is the absolute prerequisite for effective microglia/macrophage-mediated erythrophagocytosis to clear extravasated blood products during the later recovery phase.²⁷ Indeed, the failed resolution of inflammation represents a major pathological bottleneck in ICH recovery, underscoring the necessity to actively promote, rather than statically suppress, macrophage-mediated clearance mechanisms.⁷⁶ Mirroring these functional requirements, studies show that while CCL2 or CCR2 deficiency reduces initial hematoma volume, it subsequently delays hematoma clearance, indicating that CCL2-CCR2 signaling actively drives both progression and ultimate resolution.²⁶

6.2 Ischemic stroke: ischemia-reperfusion injury, BBB dysfunction, and biphasic remodeling

A defining hallmark of ischemic stroke is the secondary injury following reperfusion, which cascades through neurovascular dysfunction, massive leukocyte infiltration, oxidative stress, and structural BBB disruption.⁷⁷ Central to this pathological cascade is the CCL2-CCR2 axis, which critically modulates targeted monocyte recruitment and endothelial barrier permeability.⁷⁸ Adding structural granularity to this process, recent work in rodent models identified an astrocytic Slc4a4-CCL2 and endothelial CCR2 pathway that regulates BBB integrity and structural repair via NO-related mechanisms in both healthy and stroke brains.¹⁵ Crucially, the functional trajectory of CCR2-positive monocytes is highly stage-dependent. Far from being exclusively detrimental, the very same myeloid lineage that aggressively exacerbates acute brain injury is later implicated in driving angiogenesis and functional recovery.³⁰ Corroborating this biphasic nature, specific CCR2 deficiency in monocytes fundamentally impairs angiogenesis and delays functional recovery after ischemic stroke in mice, definitively supporting the reparative capacity of CCR2-dependent myeloid responses during the subacute phase.⁷⁹

6.3 Traumatic brain injury: persistent monocyte populations and chronic neurodegeneration

TBI presents a distinct paradigm, as its inflammatory response often persists long beyond the initial biomechanical injury, heavily contributing to delayed

neurodegenerative pathologies.²⁹ Bridging clinical observations with experimental models, robust CCL2 induction after TBI strongly correlates with sustained macrophage recruitment and chronified inflammatory responses.²⁹ In alignment with these findings, experimental Ccr2 deficiency effectively blunts macrophage infiltration and improves cognitive outcomes, underscoring the pathogenic role of CCR2-driven recruitment in post-traumatic secondary injury.⁸⁰ Shifting the paradigm further, single-cell transcriptomic analyses have revealed that severe brain injury generates immunoregulatory and neutrophil-like monocyte subsets bearing distinct transcriptional signatures, proving that post-injury monocytes are profoundly heterogeneous rather than uniformly pro-inflammatory.²⁵ Perhaps most intriguingly, in mild TBI, Ccr2 deletion reduces lesion cavity size but paradoxically enhances tau pathology, suggesting that CCR2-dependent monocyte responses exert dissociable, non-linear effects on acute structural injury versus chronic neurodegenerative endpoints.¹⁶

6.4 Spinal cord injury: myelin debris clearance, glial scar formation, and remyelination

The SCI microenvironment poses a unique set of hurdles, characterized by the physical disruption of the blood-spinal cord barrier, severe axonal tract damage, persistent accumulation of myelin debris, and subsequent fibrotic/glial scar formation.⁸¹ Driven by the excess myelin debris generated after injury, macrophage phenotypes are profoundly skewed toward lipid accumulation and foam-cell-like features, which sustain a state of secondary inflammation.³¹ Simultaneously, the evolving glial and fibrotic scar creates a double-edged sword: providing protective wound containment while erecting rigid inhibitory barriers against axonal regeneration, positioning scar biology as a major disease-specific determinant of SCI repair.⁸²

Given this immense complexity, therapeutic strategies in SCI must delicately balance inflammatory suppression with necessary debris clearance, scar remodeling, and oligodendrocyte progenitor cell-driven remyelination. Validating this approach, experimental interventions such as the administration of recombinant mouse mast cell protease 6 in rodent SCI models have successfully improved functional recovery by actively promoting remyelination and mitigating dense glial scar formation.⁷¹ Pioneering targeted precision, CCR2-engineered macrophage membrane-coated nanoparticles have recently been deployed to advance SCI repair, successfully homing to the CCL2-rich inflammatory lesion to suppress widespread neuroinflammation and neuronal death.⁵

7. Therapeutic targeting strategies

Pharmacological research targeting the CCL2-CCR2 axis is shifting from systemic receptor blockade to the exploration of interventions with greater spatiotemporal

specificity.^{2,5,18,83} Current experimental therapies attempt to utilize biased signaling modulation, local delivery platforms, and staged intervention regimens to mitigate potential adverse effects caused by systemic receptor inhibition, including impaired immune defense (Table 2).

7.1 The current landscape and translational bottlenecks of CCL2-CCR2-targeted clinical trials

To date, no published clinical trial has directly evaluated selective CCL2-targeted or CCR2-targeted therapeutics in patients with ICH or other acute CNS injuries. Consequently, human evidence in hemorrhagic stroke remains primarily confined to observational cohorts demonstrating that elevated circulating CCL2 correlates with exacerbated functional deficits.²⁰ In contrast, clinical-stage experience has been accumulated from non-CNS inflammatory, fibrotic, and oncological disorders. For instance, the oral dual CCR2/CCR5 antagonist cenicriviroc exhibited an early antifibrotic signal in the phase 2b CENTAUR study for nonalcoholic steatohepatitis, yet the subsequent phase 3 AURORA trial did not demonstrate sustained histological efficacy for liver fibrosis.⁸⁴⁻⁸⁶ Similarly, the anti-CCL2 monoclonal antibody carlumab showed limited clinical efficacy in metastatic prostate cancer and idiopathic pulmonary fibrosis, while ongoing evaluations of specific CCR2 inhibitors like PF-04136309 and plozalizumab indicate that achieving durable clinical benefit remains disease- and regimen-dependent.⁸⁷⁻⁹⁰ These clinical experiences suggest that systemic chemokine-axis blockade is pharmacologically feasible but may be insufficient without appropriate disease-specific timing and tissue-targeted delivery. For acute CNS injury, particularly ICH, this implies that static, broad-spectrum CCL2-CCR2 inhibition requires careful re-evaluation, as it must balance the mitigation of early inflammatory injury with the preservation of later macrophage-mediated hematoma clearance and tissue repair.

7.2 Drug repurposing and biased signaling modulation

Small-molecule antagonists are transitioning from simple receptor occupancy to the targeted intervention of specific downstream pathways.^{2,4,91} For instance, the dual CCR2/CCR5 antagonist cenicriviroc maintains BBB integrity in models of ischemia and human immunodeficiency virus type 1 infection by inhibiting the NLRP3 inflammasome and pyroptosis independently of its antiviral effects.^{4,92} Regarding upstream signaling modulation, the inhibition of phosphodiesterase 2A elevates intracellular cyclic nucleotide levels and promotes the phenotypic shift of microglia toward an anti-inflammatory state.⁹¹ This process reduces initial CCL2 synthesis and alleviates neuropathic pain following SCI.⁹¹ The specific CCR2 inhibitor propagermanium also exhibits anti-neuroinflammatory activity in a Tay-Sachs disease model.⁹³ Furthermore, targeting intracellular receptor-interacting proteins has achieved biased inhibition. For example, disulfiram specifically blocks the binding of CCR2 to

the cytosolic protein FROUNT. This action suppresses the excessive inflammatory migration of macrophages in an SAH model while preserving the basal phagocytic function required for hematoma clearance.²

7.3 Local competitive neutralization and biomimetic nanodelivery

Bioengineered nanoplateforms have been developed for the local competitive neutralization of chemokines to overcome the pharmacokinetic limitations of systemic antagonists. Engineered nanoparticles coated with macrophage membranes overexpressing CCR2 act as high-affinity neutralizing agents for CCL2.⁵ In SCI models, these nanoparticles sequester extracellular CCL2 and disrupt the chemotactic gradient, leading to a significant reduction in neuronal apoptosis and inflammatory infiltration.⁵ Cell-specific targeting has enabled further precision, as demonstrated by brain-targeted liposomes hybridized with M2 macrophage membranes. These liposomes deliver therapeutics directly to lesion sites and facilitate the phenotypic transition of microglia and macrophages toward a reparative state in TBI.⁹⁴ Additionally, the integration of pharmacological inhibitors with responsive polymeric carriers has advanced the management of vascular pathology, particularly in addressing the complications of reperfusion therapy. Targeted poly(lactic-co-glycolic acid) nanoparticles loaded with CCL2/CCR2 inhibitors have demonstrated the ability to attenuate hemorrhagic transformation following thrombolysis.⁹⁵ This indicates that controlled-release systems can effectively separate the therapeutic benefits of reperfusion from the detrimental impacts of inflammation-induced vascular injury, thereby providing a safe timeframe for stroke management.⁹⁵

7.4 Transcriptional silencing and extracellular vesicle therapy

RNA interference technologies offer enhanced targeting specificity. The delivery of *Ccl2* siRNA via engineered EVs effectively silences *Ccl2* transcription in reactive astrocytes.⁹⁶ This targeted delivery reduces the local recruitment of inflammatory myeloid cells and promotes motor functional recovery after SCI.⁹⁶ Moreover, exosomes derived from induced pluripotent stem cells exhibit BBB-protective efficacy in an ICH model by activating the astrocytic PI3K/AKT pathway and modulating monocyte chemoattractant protein-1 signaling.⁹⁷ These findings highlight the potential of cell-free therapies in regulating neurovascular homeostasis.⁹⁷

7.5 Staged combinatorial interventions

Recent experimental data support a staged therapeutic approach that dynamically adjusts chemokine modulation according to the temporal evolution of the injury

microenvironment. In SCI, a sequential repurposing strategy for chemokine antagonists has been proposed to address the contrasting inflammatory requirements of the acute and chronic phases. This sequential method ensures that the early suppression of neutrophil and monocyte infiltration does not impede the later stages of tissue remodeling.¹⁸ Furthermore, the integration of immunomodulation and structural support has been realized through multifaceted hydrogel systems. These nanoparticle-assembled scaffolds co-deliver anti-inflammatory agents to block the CCL2 axis while simultaneously providing a physical matrix for nerve growth, leading to significantly enhanced motor function recovery (Figure 5).⁸³

7.6 Next-generation precision immunomodulation: integrating multi-omics and AI-guided strategies

Although nanoparticle delivery systems and RNA interference provide more selective modalities for modulating the CCL2-CCR2 axis, optimizing future therapies requires precise mapping of where, when, and in which specific cell subpopulations this pathway operates. Traditional bulk transcriptomics obscures the profound spatial and cellular heterogeneity intrinsic to injured CNS tissue. To bridge this gap, recent spatial and single-cell transcriptomic profiling of traumatic and ischemic brain injuries has elucidated that glial and immune responses are intricately organized by lesion topography, post-injury timing, and cell identity.^{98,99} Similarly, emerging multi-omics and single-cell studies in ICH delineate the dynamic evolution of the immune landscape during perihematomal edema progression, highlighting that CCL2-CCR2 signaling must be contextualized within a spatially defined microenvironment.^{44,100} Furthermore, high-dimensional single-cell mass cytometry has unmasked the profound phenotypic diversity of brain myeloid cells and circulating human monocytes, extending far beyond conventional marker-based binary classifications.¹⁰¹ This granular resolution is critical because future interventions must distinguish pathogenic monocyte recruitment from reparative myeloid responses, rather than indiscriminately targeting all CCR2-positive cells as a monolithic detrimental population.

Despite these multi-omics breakthroughs, translating immense, high-dimensional datasets into actionable clinical decision-making remains in its nascent stages. Artificial intelligence (AI) and machine-learning architectures are increasingly deployed to integrate neuroimaging, fluid biomarkers, and omics datasets for the diagnosis and prognostic modeling of neurological disorders.^{102,103} Within the framework of CCL2-CCR2-targeted therapies, the most realistic near-term application of AI lies in optimizing patient stratification, forecasting individual inflammatory trajectories, and estimating personalized therapeutic windows. By continuously integrating parameters such as hematoma radiomics, fluctuating circulating chemokine levels, and real-time cell-state metadata, AI-guided algorithms can identify the precise inflection point between acute neurotoxicity and subacute resolution. Ultimately, merging high-resolution multi-omics mapping with AI-driven prediction models will propel CCL2-

CCR2 modulation away from empirical, broad-spectrum pathway inhibition toward dynamically staged, cell-state-informed precision immunomodulation (Figure 6).

8. Perspectives

The conceptual understanding of the CCL2-CCR2 axis in CNS injury has evolved from its initial characterization as a simple pro-inflammatory mediator to its current recognition as a multifaceted regulator of neurovascular homeostasis and systemic immune mobilization. The evidence synthesized in this review suggests that the limited clinical efficacy of early broad-spectrum antagonists is likely attributable to an incomplete understanding of the stage-dependent functions of chemokine signaling rather than a lack of biological potency. Non-discriminatory and prolonged receptor blockade compromises beneficial macrophage-mediated hematoma resolution and myelin debris clearance. This highlights the need for a transition toward precision modulation. Future clinical translation depends on the implementation of staged treatment regimens that dynamically align pharmacological interventions with the temporal progression of the immune response. Recent preclinical studies in SCI demonstrate that a sequential regulatory strategy, which specifically targets the infiltration of neutrophils and inflammatory monocytes during the acute phase while preserving the activity of reparative macrophages during the chronic phase, significantly enhances functional recovery compared to sustained inhibition.¹⁸

The therapeutic application of CCL2-CCR2 modulation has also extended beyond neuroprotection to encompass vascular stabilization and thrombosis management. Mechanistic studies confirm that inhibiting monocyte recruitment during the acute phase of arterial thrombosis can reduce thrombus stability and enhance fibrinolysis. This mechanism provides a strong theoretical basis for adjunctive immunomodulation in the management of acute stroke.¹⁰⁴ To optimize these interventions, patient stratification based on highly sensitive inflammatory biomarkers is crucial. Targeted proteomics has identified specific blood-borne biomarkers capable of predicting the trajectory of cerebrovascular diseases and facilitating the determination of optimal treatment timeframes.¹⁰⁵ Concurrently, single-cell transcriptomic analyses have clarified the heterogeneity of microglial repopulation by distinguishing pathogenic subpopulations from those driving tissue repair.¹⁰⁶

Translating these findings into clinical applications urgently requires the comprehensive support of multi-omics technologies. Future research must rely on single-cell transcriptomics to achieve a high-resolution characterization of the dynamic phenotypic evolution of myeloid cells at the lesion site and isolate the signature markers of pathogenic versus reparative subpopulations.²⁵ Simultaneously, integrating targeted proteomics to extensively analyze biofluid biomarkers with high temporal resolution will provide an objective basis for patient risk stratification and the definition of optimal intervention periods.¹⁰⁵ In conclusion, combining environment-responsive delivery

platforms with dynamically evolving spatiotemporal pathology to achieve a distinct separation of the pathological inflammatory cascade from physiological tissue regeneration at the molecular and cellular levels remains the essential path to overcoming current treatment limitations in severe CNS injuries and driving substantial neurofunctional recovery.

Search Methodology

To comprehensively synthesize the evidence, a literature search was conducted across PubMed, Web of Science, and Scopus databases, covering the period from inception to July 2026. The search strategy utilized both Medical Subject Headings (MeSH) and free-text terms, applying Boolean operators: (“CCL2” OR “MCP-1”) AND (“CCR2”) AND (“intracerebral hemorrhage” OR “acute CNS injury” OR “neuroinflammation” OR “macrophage polarization”). The inclusion criteria were restricted to peer-reviewed, English-language original research (both in vivo and in vitro studies) and review articles. Conference abstracts, unreviewed preprints, and non-English publications were excluded. Particular emphasis was placed on studies published within the last five years that elucidate the spatiotemporal dynamic changes of the microenvironment.

Declarations

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Consent for publication

All authors have read and approved the final version of the manuscript and consent to its submission for publication. We confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication elsewhere.

Ethical statement

Ethics approval was not required by the ethics committee, as this review article does not contain any studies with human participants or animals performed by any of the authors.

Declaration of Generative AI and AI-assisted technologies Use

During the research and preparation of this manuscript, the authors did not employ any generative AI or AI-assisted technologies for data analysis, interpretation, figure generation, or text composition. The authors are solely responsible for the integrity and originality of all content presented.

Funding

The authors' research on cerebrovascular diseases was supported by grants from the Henan Province Medical Science and Technology Research Program (SBGJ202403031); Natural Science Foundation of Henan (262300421603); Postdoctoral Fellowship Program of China Postdoctoral Science Foundation (GZC20232401); Hunan Provincial Natural Science Foundation (2023JJ40572); National Natural Science Foundation of China (U25A2065, W2541024); Key Advantageous Discipline Construction Project of Guizhou Provincial Health Commission (2023, Emergency Department); Guizhou Provincial Natural Science Foundation (Qiankehe Foundation MS[2025]432); National Natural Science Foundation of China (NSFC) Regional Fund Cultivation Program Project of the Affiliated Hospital of Guizhou Medical University (gyfynsfc[2023]-47).

CRedit authorship contribution statement

Rui Zhang: Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Liang Cao:** Conceptualization, Methodology, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing. **V. Wee Yong:** Conceptualization, Methodology, Formal analysis, Data curation, Visualization, Writing – review & editing. **Leiluo Yang:** Conceptualization, Methodology, Formal analysis, Data curation, Visualization, Writing – review & editing. **Mengzhou Xue:** Conceptualization, Methodology, Formal analysis, Data curation, Visualization, Writing – review & editing.

Data sharing statement

This article is a review, and no experimental data are available.

Acknowledgments

Not applicable.

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1 **Table 1. Temporal dynamics of CCL2-CCR2 signaling and staged therapeutic implications after acute CNS injury**

Stage	Temporal and biological features	ICH-centered evidence	Cross-disease support	Beneficial vs detrimental interpretation	Staged therapeutic implication	References
Early danger-signaling phase	Early after acute CNS injury, DAMPs, thrombin, blood-derived products, and vascular disruption initiate local chemokine responses before extensive peripheral immune-cell accumulation.	In ICH, chemokine release after hematoma formation links local inflammatory activation to peripheral immune-cell chemotaxis across the disrupted BBB.	In SCI, thrombin induces rapid astrocytic CCL2 production before later CCL5 expression and sustained macrophage/microglial infiltration.	Early CCL2 is best interpreted as an injury-alarm and immune-communication signal, not automatically as a purely detrimental pathway.	Avoid indiscriminate immediate broad blockade before the lesion context and inflammatory trajectory are clear.	10,18,19
Acute inflammatory recruitment phase	During the first days after injury, the CCL2-CCR2 axis promotes peripheral CCR2-positive inflammatory monocyte recruitment into injured CNS tissue.	In experimental ICH, blood-derived CCR2+Ly6C-high inflammatory monocytes enter the brain, outnumber other leukocyte subsets, produce TNF, and exacerbate acute neurological disability. In human ICH, circulating CCL2 is associated with poor functional outcome.	In ischemic stroke, peripheral monocytes can infiltrate lesions within 24 h, peak between 3 and 7 days, and differentiate into macrophages.	Excessive early CCR2-positive monocyte influx is mainly injury-amplifying, although limited myeloid recruitment may initiate debris handling.	Short-window inhibition of excessive CCR2-positive inflammatory monocyte recruitment may be beneficial during acute perihematomal inflammation.	20-22

Neurovascular injury and BBB-disruption phase	Chemokine signaling and inflammatory mediators can aggravate BBB disruption, edema, and secondary injury.	In acute ICH, S1PR3 is implicated in BBB injury through the CCL2-CCR2 axis; ICH-associated ROS-NLRP3 signaling reinforces oxidative-inflammatory secondary injury.	In ischemic stroke, astrocytic Slc4a4 deletion increases CCL2 secretion and worsens BBB disruption through a CCL2-CCR2/NO-related mechanism.	CCL2-CCR2 activity at the neurovascular interface may amplify BBB injury when excessive, while tightly regulated neurovascular immune signaling may participate in later repair.	Prefer local, time-limited, or pathway-specific modulation to reduce BBB injury while avoiding prolonged systemic CCR2 suppression.	15,23,24
Subacute hematoma/debris clearance phase	After the early inflammatory phase, recruited myeloid cells increasingly participate in hematoma resolution, erythrophagocytosis, debris handling, and tissue remodeling.	In ICH, CCL2 or CCR2 deficiency reduces early hematoma volume but delays subsequent hematoma recovery, indicating dual participation in hematoma progression and clearance. Macrophage erythrophagocytosis accelerates hematoma resolution and improves neurological recovery.	Across acute CNS injuries, macrophage phenotypes change with local cues and time, making late myeloid responses distinct from early inflammatory recruitment.	The same axis that aggravates early inflammation can support later clearance and repair-associated macrophage functions.	Avoid prolonged, non-selective CCR2 blockade once hematoma clearance and tissue repair become important.	26,27
Repair-associated immune heterogeneity phase	Post-injury monocyte/macrophage responses are heterogeneous rather than fixed in a single pro-inflammatory phenotype.	ICH evidence showing delayed hematoma recovery after CCL2 or CCR2 deficiency supports recovery-associated roles of recruited myeloid cells.	Single-cell transcriptomic analysis after brain injury identifies immunoregulatory and neutrophil-like monocyte subsets. Ischemic stroke studies show that CCR2-positive monocytes can contribute to	The beneficial or detrimental outcome depends on timing, cell state, and disease context rather than receptor activation alone.	Shift from broad inhibition to phenotype-preserving or phenotype-redirecting immunomodulation.	25,26,79

	angiogenesis and functional recovery during repair.					
Later recovery / chronic remodeling phase	Later responses involve glial-myeloid interactions, scar organization, tissue stabilization, and potential maladaptive chronic inflammation.	In ICH, stage-dependent microglia-astrocyte interaction contributes to neuroprotective glial scar formation, indicating that later glial responses are not uniformly detrimental.	In TBI, CCL2 elevation can persist beyond the immediate post-injury period, and Ccr2 deletion dissociates lesion cavity formation from tau pathology.	Persistent chemokine signaling may be maladaptive, but coordinated glial-myeloid responses can contribute to structural containment and repair.	Suppress unresolved maladaptive inflammation while preserving repair-supportive glial-myeloid interactions.	16,28,29

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5 Table 2. Preclinical therapeutic approaches related to CCL2-CCR2-targeted immunomodulation in acute CNS injury models

Strategy/platform	Target or pathway	Disease/model	Evidence type/species	Main experimental findings	Translational consideration	Reference
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Disulfiram-mediated FROUNT inhibition	CCR2/CCR5-associated FROUNT signaling	Subarachnoid hemorrhage (SAH)	Rodent experimental study; hemorrhagic CNS injury model	Suppressed excessive macrophage migration after SAH while aiming to preserve phagocytic functions relevant to hemorrhage resolution.	Potentially more selective than broad CCR2 blockade, but CNS injury evidence remains preclinical and mainly hemorrhage-model based.	2
Dual CCR2/CCR5 antagonist cenicriviroc	CCR2/CCR5; NLRP3 inflammasome and pyroptosis	Ischemia/HIV-1-associated BBB injury model	Cell and experimental disease-model evidence	Preserved astrocyte-endothelial communication and reduced BBB-related inflammatory injury through inhibition of NLRP3 inflammasome and pyroptosis.	Supports BBB-protective potential, but direct evidence in ICH remains limited.	4
CCL2/CCR2 inhibitor-loaded targeted PLGA nanoparticles	CCL2/CCR2 axis	Post-thrombotic hemorrhagic transformation after ischemic stroke	Rodent experimental study; reperfusion-associated hemorrhagic transformation	Targeted nanoparticle delivery of a CCL2/CCR2 inhibitor attenuated hemorrhagic transformation after thrombolysis and improved neurological recovery.	Relevant to vascular injury after reperfusion; should be distinguished from primary spontaneous ICH.	95
CCR2-overexpressing macrophage membrane-coated nanoparticles	Extracellular CCL2 neutralization	Spinal cord injury (SCI)	Rodent experimental study	CCR2-enriched macrophage membrane coating sequestered CCL2 in the inflammatory lesion microenvironment, reduced inflammatory infiltration and neuronal death, and promoted functional recovery.	Provides a local chemokine-decoy strategy, but long-term safety and dosing windows require validation.	5

Engineered extracellular vesicles carrying Ccl2 siRNA	Astrocytic Ccl2 transcription	Traumatic spinal cord injury	Rodent experimental study	Engineered EVs delivered Ccl2 siRNA to reactive astrocytes, reduced inflammatory myeloid-cell recruitment, and improved motor recovery after SCI.	Cell-source specificity is a strength, but delivery efficiency and off-target effects need further study.	96
hiPSC-derived neural stem cell exosomes	PI3K/AKT/MCP-1 signaling in astrocytes	Intracerebral hemorrhage (ICH)	Rodent experimental study; human cell-derived exosome product	hiPSC-NSC-derived exosomes improved BBB function after ICH by activating astrocytic PI3K/AKT signaling and modulating MCP-1-related responses.	Particularly relevant to Brain Hemorrhages, but the intervention is not a direct CCR2 antagonist and requires mechanistic clarification.	97
M2 macrophage membrane-hybrid biomimetic liposomes	Lesion-targeted anti-inflammatory delivery	Traumatic brain injury (TBI)	Rodent experimental study	Brain-targeted biomimetic liposomes reduced inflammatory responses and promoted a more reparative microenvironment after TBI.	Useful as a biomimetic delivery example, but not a direct CCL2-CCR2 inhibitor.	94
Phase-resolved chemokine antagonist combination	Sequential chemokine modulation	Spinal cord injury (SCI)	Rodent experimental study; therapeutic repurposing framework	Combined and stage-adjusted chemokine antagonist treatment was proposed to match the changing inflammatory requirements after SCI.	Strongly supports staged therapy, but requires validation across hemorrhagic and ischemic CNS injury models.	18
Nanoparticle-assembled multifaceted hydrogel	Local anti-inflammatory delivery and structural support	Spinal cord injury (SCI)	Rodent experimental study	Hydrogel-based therapy combined inflammatory modulation with structural	Demonstrates combined immunomodulatory and regenerative	83

				support for nerve repair and improved functional recovery after SCI.	delivery, but CCL2-axis specificity should be stated cautiously.	
PDE2A inhibition	cAMP/cGMP-associated microglial modulation and CCL2 reduction	SCI-related neuropathic pain model	Rodent experimental study	PDE2A inhibition reduced CCL2 synthesis, shifted microglial responses toward an anti-inflammatory state, and relieved mechanical allodynia after SCI.	More relevant to chronic pain after SCI than to acute structural CNS injury; should be presented as supportive rather than central evidence.	⁹¹

Figure legends

Figure 1. Systemic mobilization of peripheral inflammatory monocytes driven by DAMP-induced amplification of the CCL2 signaling cascade following CNS injury.

Following acute CNS injury, DAMP-activated glial cells synthesize massive amounts of CCL2 via NF- κ B and NLRP3 pathways. This locally produced CCL2 leaks into the systemic circulation, establishing a chemotactic gradient. In the hematopoietic niche, circulating CCL2 binds to CCR2 on Ly6C inflammatory monocytes. This interaction directly antagonizes CXCR4-CXCL12 homeostatic retention signals, triggering the systemic mobilization and targeted chemotaxis of these monocytes into the damaged CNS parenchyma. Abbreviations: ATP, adenosine triphosphate; BBB, blood-brain barrier; CCL2, C-C motif chemokine ligand 2; CCR2, C-C motif chemokine receptor 2; CNS, central nervous system; CXCL12, C-X-C motif chemokine ligand 12; CXCR4, C-X-C motif chemokine receptor 4; DAMPs, damage-associated molecular patterns; HMGN1, high mobility group nucleosome binding protein 1; NF- κ B, nuclear factor kappa B; NLRP3, NLR family pyrin domain containing 3; PAR-1, protease-activated receptor 1; P2X7, P2X purinoceptor 7; TLR4, Toll-like receptor 4.

Figure 2. CCL2-CCR2 axis-mediated regulation of endothelial tight junctions and alterations in blood-brain barrier permeability.

CCL2 binding to inducibly expressed CCR2 on brain microvascular endothelial cells (BMECs) induces p38 MAPK phosphorylation, triggering the downregulation of tight junction proteins (Occludin and Claudin-5).^{4,15,78} Concurrently, a positive feedback loop emerges: Slc4a4-mediated intracellular alkalinization in reactive astrocytes drives massive paracrine CCL2 secretion. This astrocyte-derived CCL2 further exacerbates tight junction disassembly via a nitric oxide-dependent mechanism. These synergistic mechanisms widen the paracellular gap, resulting in vasogenic edema and facilitating leukocyte transmigration into the CNS. Abbreviations: BMECs, brain microvascular endothelial cells; CCL2, C-C motif chemokine ligand 2; CCR2, C-C motif chemokine receptor 2; CNS, central nervous system; MAPK, mitogen-activated protein kinase; Slc4a4, solute carrier family 4 member 4.

Figure 3. Interactive signaling and inflammatory phenotypic polarization between infiltrating myeloid cells and resident glia.

Recruited CCR2⁺ monocytes differentiate into pro-inflammatory monocyte-derived macrophages (MoDMs) that secrete cytokines including TNF- α , IL-1 β , and Lcn2. Lcn2 binding to astrocytic Lcn2R drives their transition into reactive astrocytes via

JAK2/STAT3 and NF- κ B pathways. These reactive astrocytes upregulate complement C3 and augment CCL2 synthesis. This reciprocal intercellular crosstalk creates a pathological positive feedback loop that sustains microenvironmental inflammation, promotes further myeloid recruitment and pro-inflammatory activation, and actively suppresses the transition toward reparative phenotypes, ultimately hindering tissue remodeling. Abbreviations: CCL2, C-C motif chemokine ligand 2; CCR2, C-C motif chemokine receptor 2; IL-1 β , interleukin-1 beta; JAK2, Janus kinase 2; Lcn2, lipocalin-2; Lcn2R, lipocalin-2 receptor; MoDMs, monocyte-derived macrophages; NF- κ B, nuclear factor kappa B; STAT3, signal transducer and activator of transcription 3; TNF- α , tumor necrosis factor-alpha.

Figure 4. Mechanisms of neuronal injury mediated by excitotoxicity, synaptic phagocytosis, and oxidative stress.

CCL2 binding to postsynaptic CCR2 induces excitotoxicity by increasing GluA1-AMPA receptor trafficking and enhancing GluN2B-NMDA receptor activity, which amplifies calcium (Ca^{2+}) influx. Concurrently, complement C3 deposition tags synapses for microglial clearance via CR3. Furthermore, CCL2-CCR2 signaling drives redox imbalance by activating NOX2 to generate ROS while inhibiting the Nrf2-GPX4 antioxidant pathway. During hemorrhage-induced iron (Fe^{2+}) overload, this profound redox failure catalyzes lipid peroxidation via the Fenton reaction, culminating in neuronal ferroptosis. Abbreviations: AMPA, α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid; C3, complement component 3; CCL2, C-C motif chemokine ligand 2; CCR2, C-C motif chemokine receptor 2; CR3, complement receptor 3; GluA1, AMPA receptor subunit; GluN2B, NMDA receptor subunit; GPX4, glutathione peroxidase 4; NMDA, N-methyl-D-aspartate; NOX2, NADPH oxidase 2; Nrf2, nuclear factor erythroid 2-related factor 2; ROS, reactive oxygen species.

Figure 5. Multidimensional precision intervention strategies targeting the CCL2-CCR2 axis: extracellular neutralization, intracellular biased modulation, and transcriptional silencing.

(1) **Extracellular neutralization:** Macrophage membrane-coated nanoparticles act as local decoys, specifically sequestering extracellular CCL2 to block chemotactic gradients without disrupting systemic immunity. (2) **Intracellular biased signaling:** Small-molecule modulators (e.g., disulfiram) selectively decouple CCR2 from the cytosolic protein FROUNT. This halts pathogenic migration while preserving essential macrophage clearance functions. (3) **Transcriptional silencing:** During subacute/chronic phases, engineered EVs deliver siRNA to silence astrocytic CCL2 transcription, resolving chronic inflammation and promoting tissue remyelination. Abbreviations: CCL2, C-C motif chemokine ligand 2; CCR2, C-C motif chemokine receptor 2; CNS, central nervous system; EVs, extracellular vesicles; FROUNT,

fibronectin type III domain-containing protein; NLRP3, NLR family pyrin domain containing 3; siRNA, small interfering RNA.

Figure 6. Spatiotemporal evolution of CCL2-CCR2 signaling and stage-specific immunomodulatory strategies in acute CNS injury.

Acute CCL2 upregulation drives pro-inflammatory Ly6^{Chi}/CCR2⁺ monocyte recruitment, exacerbating disease-specific secondary injuries (e.g., ICH-induced ferroptosis, stroke-related barrier leakage, SCI myelin accumulation). During the subacute phase, the microenvironment promotes myeloid transition toward reparative macrophages essential for debris clearance and inflammation resolution, dictating adaptive versus maladaptive outcomes. Consequently, interventions must be stage-specific: early therapies should restrict pathogenic influx, whereas delayed modulation must preserve clearance. Integrating spatial multi-omics and machine learning facilitates identifying precise pathological inflection points for individualized intervention. Abbreviations: BBB, blood-brain barrier; CCL2, C-C motif chemokine ligand 2; CCR2, C-C motif chemokine receptor 2; EV, extracellular vesicle; Fe²⁺, ferrous iron; I/R, ischemia-reperfusion; siRNA, small interfering RNA.

Figure 1

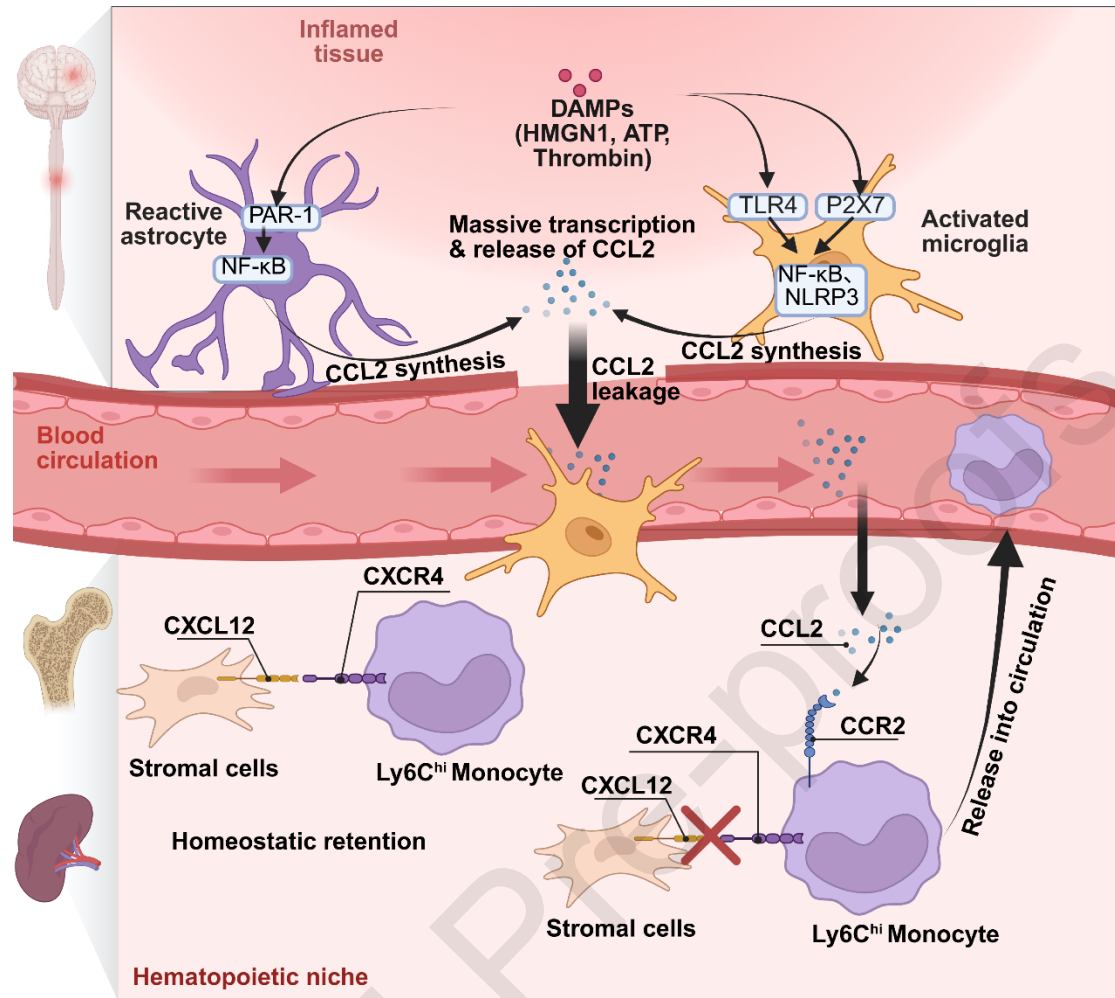


Figure 2

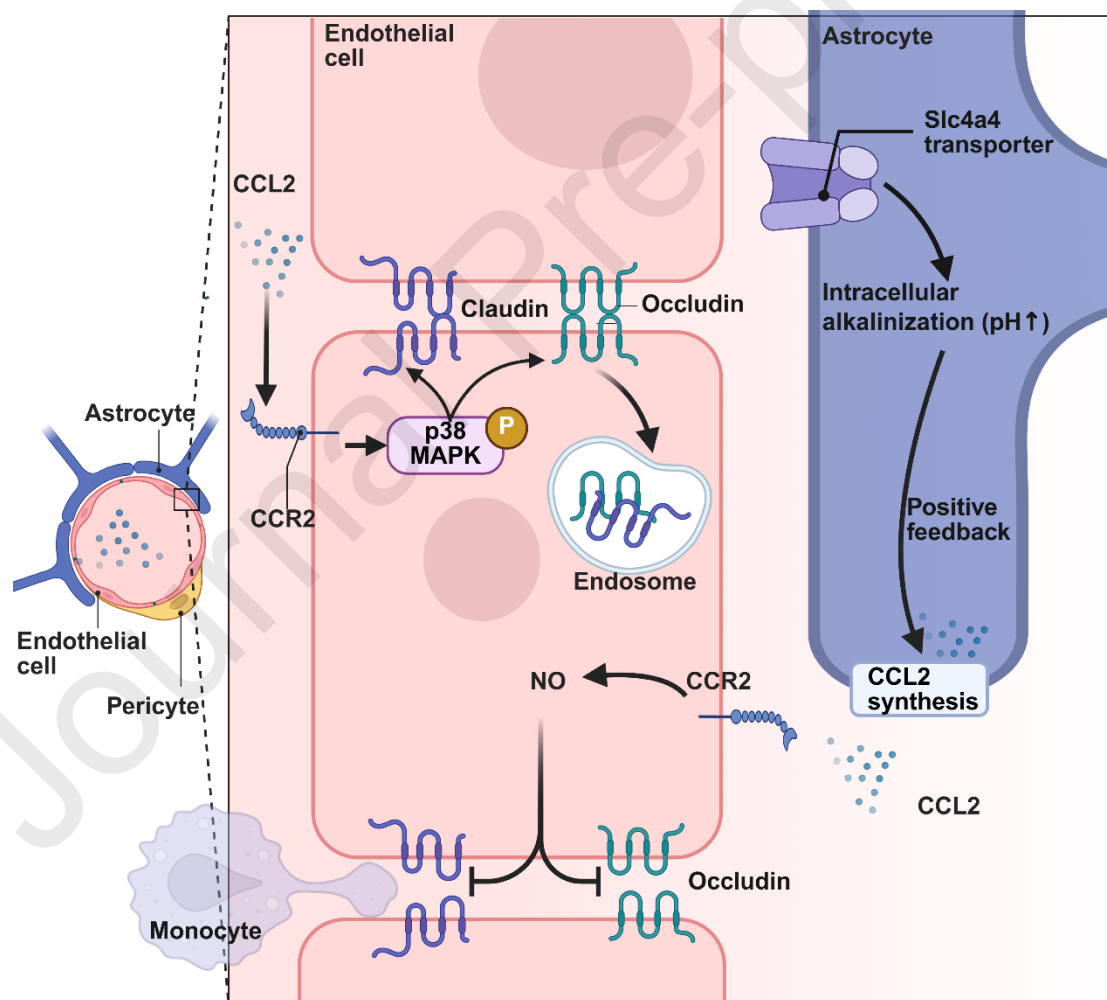


Figure 3

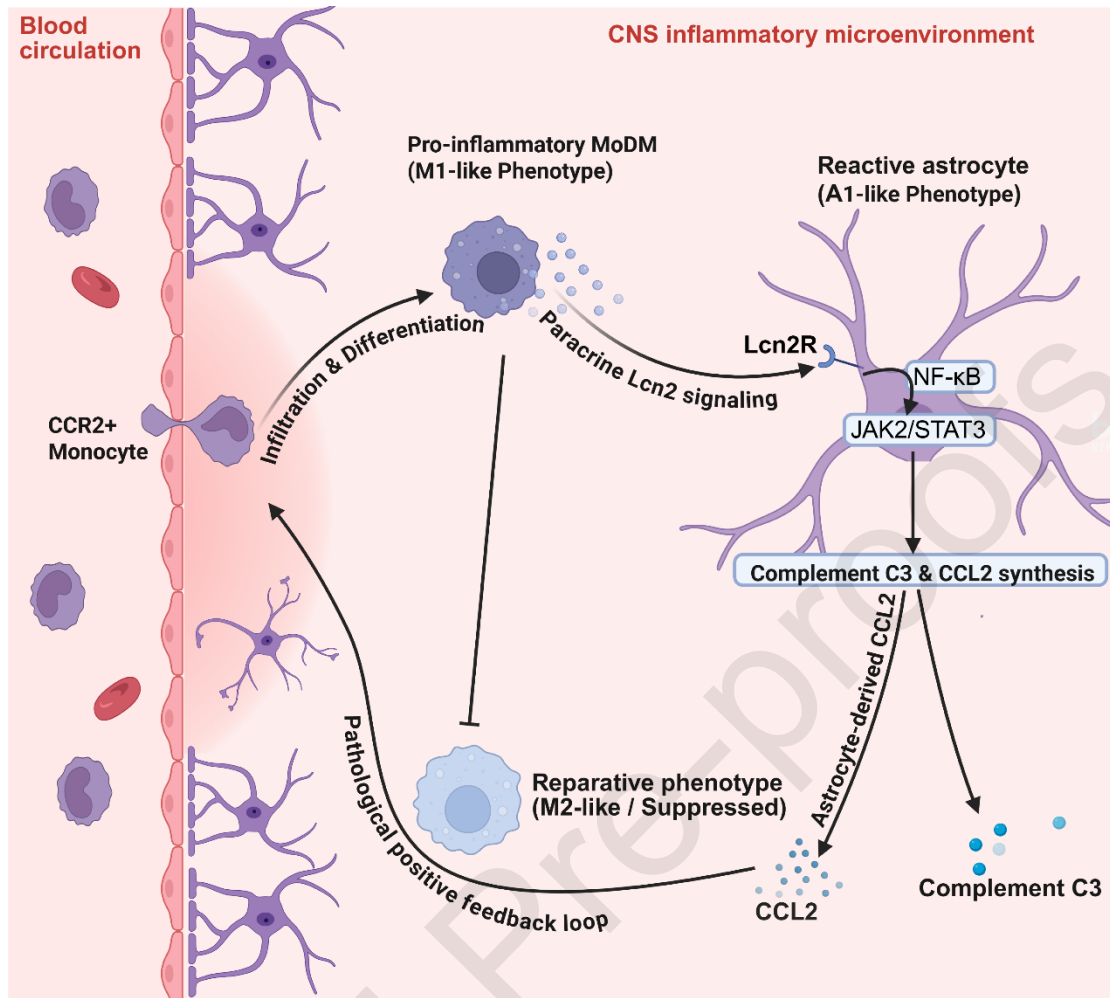


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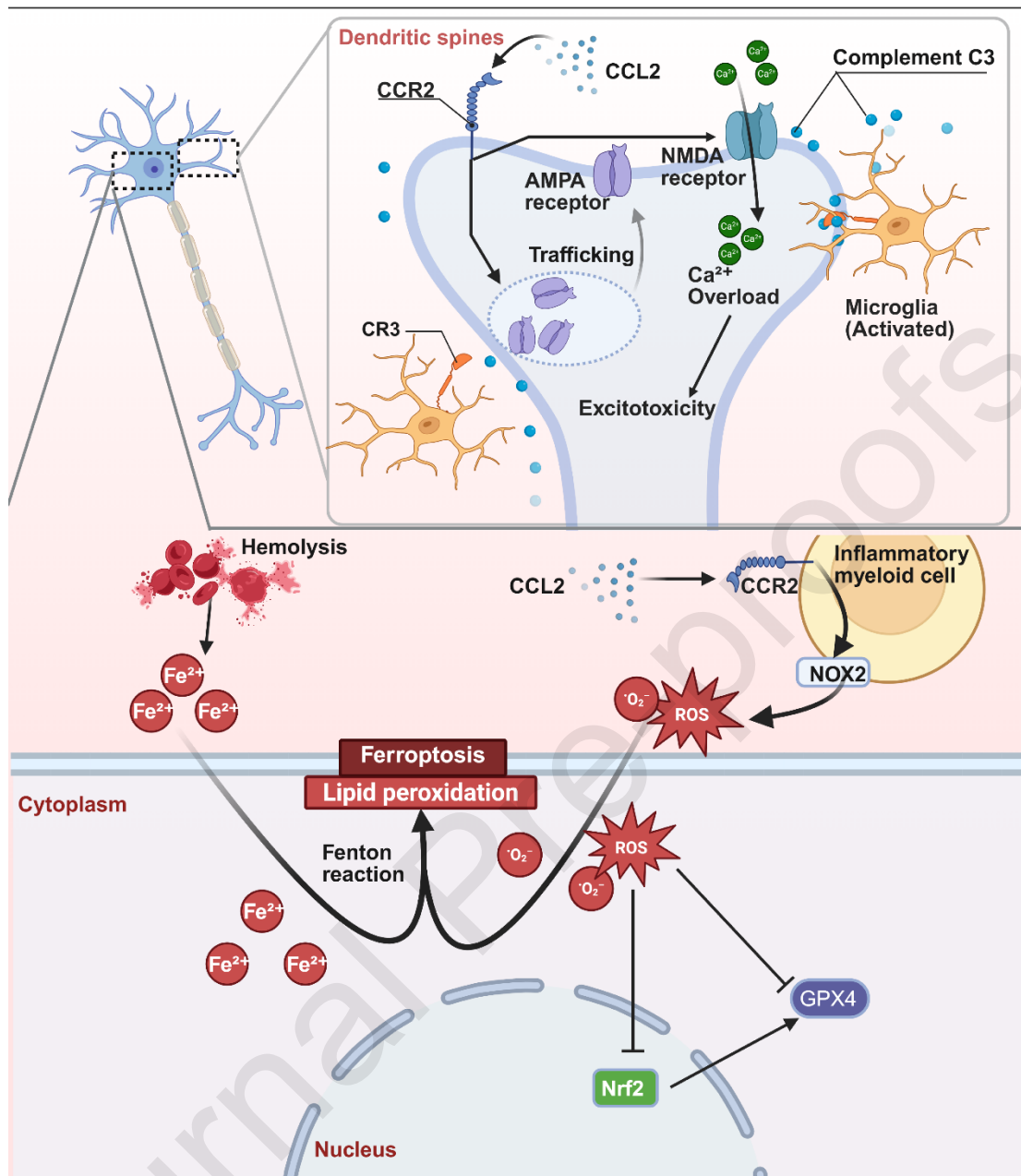


Figure 5

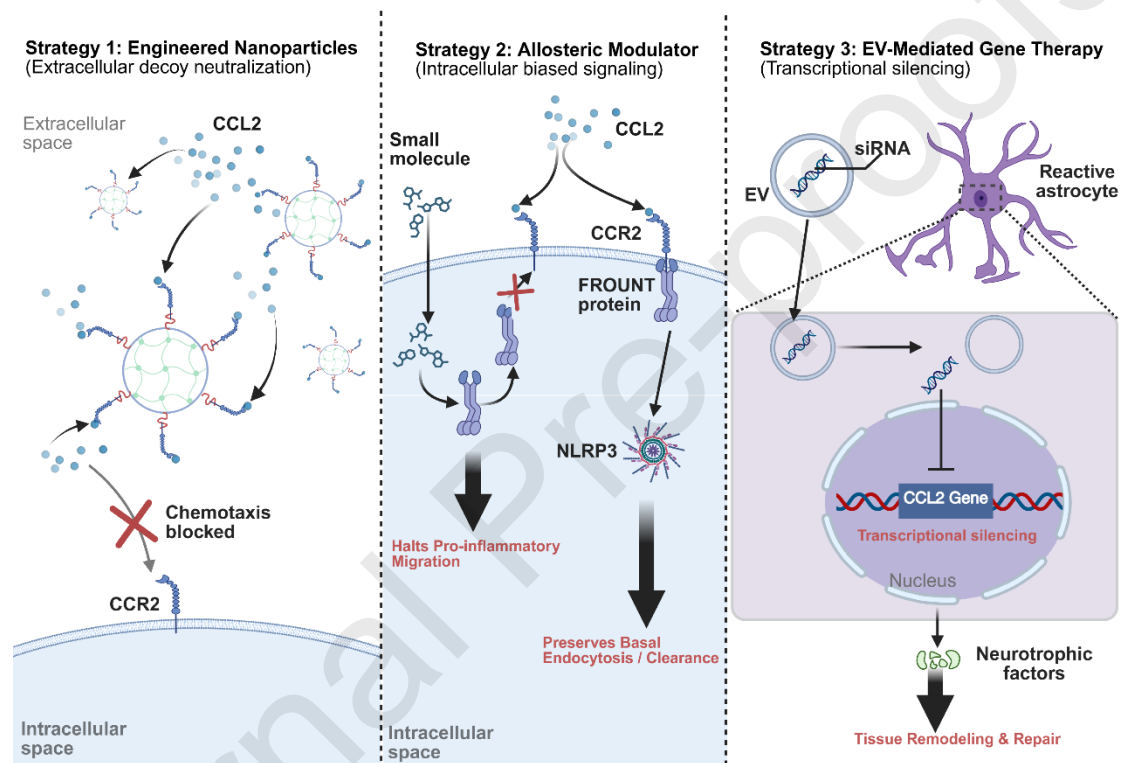
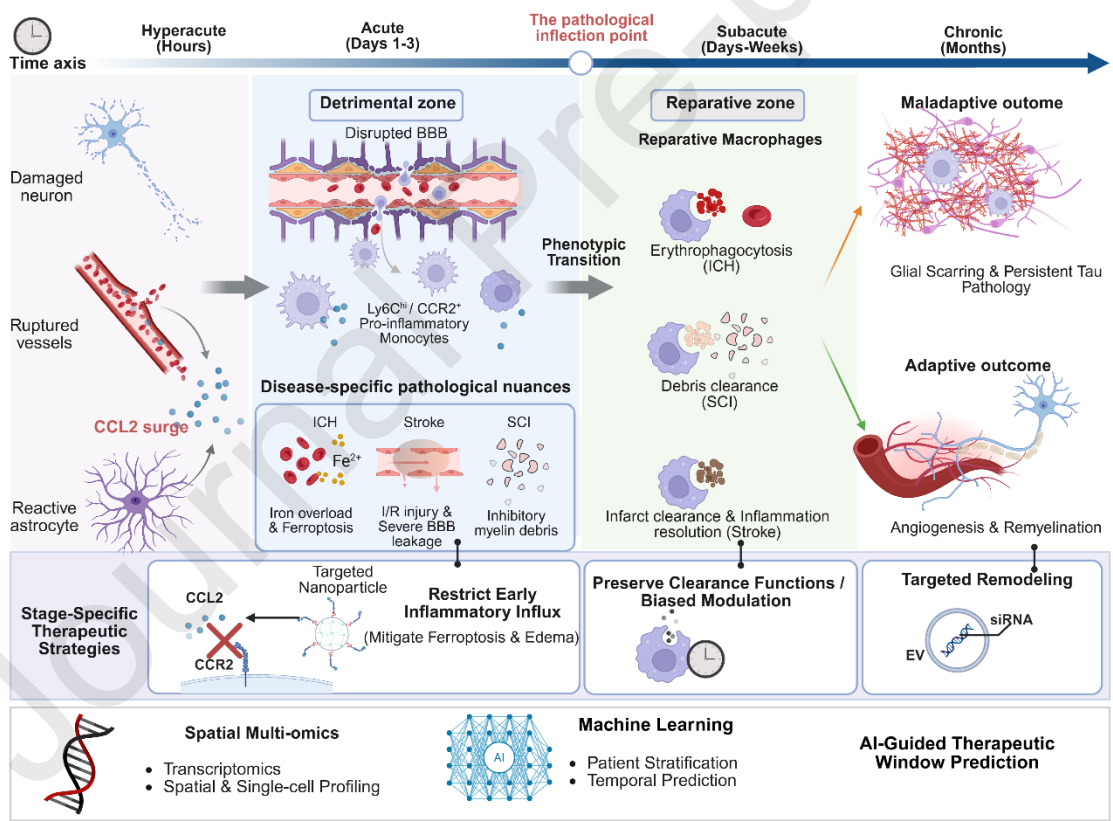


Figure 6



Not applicable.